

Lessons

- Table of Contents
- Confidence Levels
- Notes
- Bookmarks
- Highlights

Solving Multistep Equations

# Solving Multistep Equations

Some problems require more than one inverse operation to find a solution. When an equation uses more than one operation, reverse the order of operations. In most situations, undo addition or subtraction first. Then undo multiplication or division.

An equation is like a balance scale. If you perform the exact same operation on both sides of the equation, the two sides will remain balanced. The balance scale shows the equation  $3x + 5 = 26$ .

To solve, first subtract 5 from both sides of the equation.

$3x + 5 - 5 = 3x$

$26 - 5 = 21$

Next, divide both sides by 3.

$\frac{3x}{3} = x$

$\frac{21}{3} = 7$

The value of  $x$  is 7.

Remember, the goal in solving an equation is to isolate the variable on one side of the equation. Keep the equation balanced by always performing the same operation on both sides of the equation.



Evaluating/Expanding  
BEDMAS

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SAMDEB

$3x = \frac{21}{3}$

$x = 7$

Example

Hide

Solve for  $x$ :  $9 = \frac{x}{4} + 7$

Step 1

Since 7 is added on the right side of the equation, subtract 7 from both sides.

$9 = \frac{x}{4} + 7$

$9 - 7 = \frac{x}{4} + 7 - 7$

$9 - 7 = \frac{x}{4} + 7 - 7$

$2 = \frac{x}{4}$

Step 2

The variable is divided by 4, so multiply both sides by 4.

$2 = \frac{x}{4}$

$4 \cdot 2 = \frac{x}{4} \cdot 4$

$8 = x$

$2(4) = \left(\frac{x}{4}\right)(4)$

$8 = x$

$8 = x$  is the same thing as  $x = 8$ . The solution is 8.

Check:

$9 = \frac{8}{4} + 7$

$9 = 2 + 7$

$9 = 9$

$LHS = 9$

$RHS = \frac{8}{4} + 7$

$= 2 + 7$

$= 9$

$= LHS \checkmark$

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Category

Lesson 3: Solving Equations

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